

MH2100 Calculus III

Midterm Examination (Solutions)

AY 2025/2026, Semester 1

Quantitative Research Society @NTU

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Overview

- **Coverage:** limits and total differentiability; directional derivatives and the chain rule; extrema and Lagrange multipliers; double integrals and polar coordinates.
- **Format:** four problems, 100 marks in total; 110 minutes.
- **Source:** MH2100 Midterm Examination, AY2025/2026 Semester 1. The questions below reproduce the paper; the solutions and marking schemes are unofficial QRS@NTU material.

Instructions

Answer **all** questions. Show all working clearly; answers without justification may receive zero marks.

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Question 1**(25 Marks)****PROBLEM 1.**

(a) Evaluate

$$\lim_{(x,y) \rightarrow (2,1)} \frac{x^2y - x^2 - 4y + 4}{xy - x - 2y + 2}. \quad (5 \text{ points})$$

(b) Determine whether or not the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, defined by

$$\forall x, y \in \mathbb{R} : \quad f(x, y) \stackrel{\text{df}}{=} \sqrt[3]{x^3 + y^3},$$

is totally differentiable at $(0, 0)$.**(20 points)****Solution****Method 1: Direct computation**

(a) Factor both numerator and denominator:

$$x^2y - x^2 - 4y + 4 = (y - 1)(x^2 - 4) = (y - 1)(x - 2)(x + 2),$$

$$xy - x - 2y + 2 = (x - 2)(y - 1).$$

Thus, wherever the displayed quotient is defined and $(x, y) \neq (2, 1)$,

$$\frac{x^2y - x^2 - 4y + 4}{xy - x - 2y + 2} = x + 2.$$

Therefore

$$\boxed{\lim_{(x,y) \rightarrow (2,1)} \frac{x^2y - x^2 - 4y + 4}{xy - x - 2y + 2} = 4.}$$

(b) First calculate the partial derivatives at the origin. Since real cube roots preserve sign,

$$f_x(0, 0) = \lim_{h \rightarrow 0} \frac{\sqrt[3]{h^3} - 0}{h} = 1, \quad f_y(0, 0) = \lim_{h \rightarrow 0} \frac{\sqrt[3]{h^3} - 0}{h} = 1.$$

If f were totally differentiable at $(0, 0)$, its derivative would be

$$L(h, k) = h + k.$$

Approach the origin along $(h, k) = (t, t)$. Then

$$f(t, t) = \sqrt[3]{2t^3} = \sqrt[3]{2}t, \quad L(t, t) = 2t.$$

Consequently,

$$\frac{f(t, t) - f(0, 0) - L(t, t)}{\sqrt{t^2 + t^2}} = \frac{(\sqrt[3]{2} - 2)t}{\sqrt{2}|t|},$$

which has no limit 0 as $t \rightarrow 0$. Hence the defining remainder condition for total differentiability fails. Thus

$$\boxed{f \text{ is not totally differentiable at } (0, 0).}$$

Marking scheme (25 marks)**(a) (5 marks)**

- 2 marks: factors the numerator correctly.
- 1 mark: factors the denominator correctly.
- 1 mark: cancels the common factors on the domain of the quotient.
- 1 mark: obtains the limit 4.

(b) (20 marks)

- 4 marks: computes $f_x(0,0) = f_y(0,0) = 1$.
- 4 marks: identifies the only possible derivative $L(h,k) = h + k$.
- 6 marks: chooses and evaluates a valid path, such as (t, t) .
- 4 marks: shows that the normalized remainder does not tend to zero.
- 2 marks: states the correct conclusion about total differentiability.

Question 2**(25 Marks)****PROBLEM 2.**

- (a) A function
- $f : \mathbb{R}^3 \rightarrow \mathbb{R}$
- is defined by

$$\forall x, y, z \in \mathbb{R} : \quad f(x, y, z) \stackrel{\text{df}}{=} z^2 - x^3 y^2.$$

Let $\mathbf{u}$ denote the unit vector pointing from $(1, -1, 1)$ toward $(2, 1, 0)$.Compute $(D_{\mathbf{u}}f)(1, -1, 1)$ – with justification.**(10 points)**

- (b) Two functions,
- $f, g : \mathbb{R}^2 \rightarrow \mathbb{R}$
- , satisfy the following four properties:

- f is totally differentiable at $(1, 4)$.
- The tangent plane to the graph of f at $(1, 4, f(1, 4))$ is

$$\{(x, y, z) \in \mathbb{R}^3 \mid 3x + 5y = z + 26\}.$$

- $f_{xy}(1, 4) = f_{yx}(1, 4) = 1$ and $f_{xx}(1, 4) = f_{yy}(1, 4) = -1$.
- For all $x, y \in \mathbb{R}$, we have $g(x, y) = x \cdot f(x^2 - y, x + 2y)$.

Compute $g_{xy}(-2, 3)$ – with justification.**(15 points)****Solution****Method 1: Regularity and the intended chain-rule calculation**

- (a) The displacement vector is

$$(2, 1, 0) - (1, -1, 1) = (1, 2, -1),$$

whose length is $\sqrt{6}$. Hence

$$\mathbf{u} = \frac{1}{\sqrt{6}}(1, 2, -1).$$

Also,

$$\text{grad } f(x, y, z) = (-3x^2 y^2, -2x^3 y, 2z), \quad \text{grad } f(1, -1, 1) = (-3, 2, 2).$$

Since f is differentiable,

$$(D_{\mathbf{u}}f)(1, -1, 1) = \text{grad } f(1, -1, 1) \cdot \mathbf{u} = \frac{-3 + 4 - 2}{\sqrt{6}}.$$

Therefore

$$\boxed{(D_{\mathbf{u}}f)(1, -1, 1) = -\frac{1}{\sqrt{6}}}.$$

- (b) Write

$$u = x^2 - y, \quad v = x + 2y, \quad g(x, y) = xf(u, v).$$

The given point maps to $(u, v) = (1, 4)$ when $(x, y) = (-2, 3)$. From the tangent plane,

$$z = 3x + 5y - 26,$$

so

$$f_x(1, 4) = 3, \quad f_y(1, 4) = 5.$$

Regularity point. With the convention $g_{xy} = (g_x)_y$, the four properties in the question do *not* determine $g_{xy}(-2, 3)$. They give the four iterated partial derivatives of f at one point, but do not say that f_x and f_y are differentiable in a neighbourhood of that point. Thus the second-order chain rule used below is not justified by the literal hypotheses.

Under the natural additional assumption that f is C^2 on a neighbourhood of $(1, 4)$, the intended calculation is as follows. Differentiating first with respect to x gives

$$g_x = f(u, v) + x[2xf_x(u, v) + f_y(u, v)].$$

Set

$$H = -f_x + 2f_y, \quad K = 2xf_x + f_y,$$

where the derivatives of f are evaluated at (u, v) . Then

$$g_{xy} = H + xK_y,$$

and, since $u_y = -1$ and $v_y = 2$,

$$K_y = 2x(-f_{xx} + 2f_{xy}) + (-f_{yx} + 2f_{yy}).$$

At $(x, y) = (-2, 3)$, this gives

$$H(1, 4) = -3 + 2(5) = 7,$$

$$K_y = (-4)(-(-1) + 2 \cdot 1) + (-1 + 2(-1)) = -12 - 3 = -15.$$

Hence the intended, C^2 -conditional answer is

$$g_{xy}(-2, 3) = 7 + (-2)(-15), \quad \boxed{g_{xy}(-2, 3) = 37 \text{ if } f \text{ is } C^2 \text{ near } (1, 4).}$$

For completeness, the missing hypothesis is substantive. Let $U = s - 1$, $V = t - 4$, and define

$$F_0(s, t) = -3 + 3U + 5V - \frac{1}{2}U^2 + UV - \frac{1}{2}V^2,$$

$$Q(U, V) = \begin{cases} \frac{U^2V^2}{U^2 + V^2}, & (U, V) \neq (0, 0), \\ 0, & (U, V) = (0, 0). \end{cases} \quad F = F_0 + Q.$$

The function Q is C^1 , its first derivatives vanish at $(0, 0)$, and its four iterated coordinate partial derivatives there are all 0. Thus F satisfies every stated datum for f . However, along $x = -2$, $y = 3 + t$, we have $(U, V) = (-t, 2t)$, $Q_U(-t, 2t) = -32t/25$, and $Q_V(-t, 2t) = 4t/25$. The contribution of Q to $(g_x)_y(-2, 3)$ is therefore

$$(-2) \left[(-4) \left(-\frac{32}{25} \right) + \frac{4}{25} \right] = -\frac{264}{25}.$$

For this admissible F , $g_{xy}(-2, 3) = 37 - 264/25 = 661/25$, rather than 37. Consequently, under the literal statement the requested value is underdetermined.

Marking scheme (25 marks)**(a) (10 marks)**

- 3 marks: finds and normalizes the direction vector.
- 4 marks: computes $\text{grad } f(1, -1, 1) = (-3, 2, 2)$.
- 2 marks: applies the directional-derivative formula with justification.
- 1 mark: obtains $-1/\sqrt{6}$.

(b) (15 marks)

- 3 marks: extracts $f_x(1, 4) = 3$ and $f_y(1, 4) = 5$ from the tangent plane.
- 2 marks: identifies $(x^2 - y, x + 2y) = (1, 4)$ at $(-2, 3)$.
- 3 marks: explains that the stated data do not justify the second-order chain rule.
- 4 marks: gives a valid counterexample or otherwise establishes that the literal question is underdetermined.
- 3 marks: under an explicit C^2 -neighbourhood assumption, differentiates g_x , applies the chain rule correctly, and obtains 37.

Question 3**(25 Marks)****PROBLEM 3.**

A function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ is defined by

$$\forall x, y \in \mathbb{R} : \quad f(x, y) \stackrel{\text{df}}{=} e^{-xy}.$$

Let $C = \{(x, y) \in \mathbb{R}^2 \mid x^2 + 4y^2 \leq 1\}$.

- (a) Determine all the critical point(s) of $f|_C$. **(5 points)**

Note: Every critical point of $f|_C$ belongs – by definition – to the interior of C .

- (b) Determine the nature of each critical point of $f|_C$. **(5 points)**

Note: Apply the Second-Derivative Test.

- (c) Solve the Lagrange-multiplier problem for minimizing/maximizing f on

$$\text{Bd}(C) = \{(x, y) \in \mathbb{R}^2 \mid x^2 + 4y^2 = 1\},$$

which is the boundary of C in $\mathbb{R}^2$.

(10 points)

Note: You are required to apply the Method of Lagrange Multipliers.

- (d) Explain why $f|_C$ attains an absolute minimum/maximum. **(3 points)**

- (e) Determine the absolute minimum and absolute maximum of $f|_C$. **(2 points)**

Solution**Method 1: Interior analysis and Lagrange multipliers**

- (a) The gradient is

$$\text{grad } f(x, y) = e^{-xy}(-y, -x).$$

Because $e^{-xy} > 0$, both components vanish exactly when $x = y = 0$. This point belongs to the interior of C , so

The only critical point of $f|_C$ is $(0, 0)$.

- (b) The second derivatives are

$$f_{xx} = y^2 e^{-xy}, \quad f_{yy} = x^2 e^{-xy}, \quad f_{xy} = (xy - 1)e^{-xy}.$$

At $(0, 0)$,

$$f_{xx}(0, 0) = 0, \quad f_{yy}(0, 0) = 0, \quad f_{xy}(0, 0) = -1.$$

Thus the second-derivative discriminant is

$$D = f_{xx}f_{yy} - f_{xy}^2 = 0 \cdot 0 - (-1)^2 = -1 < 0.$$

Therefore

$(0, 0)$ is a saddle point of $f|_C$.

(c) Let $h(x, y) = x^2 + 4y^2$. On $\text{Bd}(C)$, the Lagrange equations are

$$-ye^{-xy} = 2\lambda x, \quad -xe^{-xy} = 8\lambda y, \quad x^2 + 4y^2 = 1.$$

Neither x nor y can be zero in the first two equations together with the constraint. Dividing the two equations yields

$$\frac{y}{x} = \frac{x}{4y}, \quad \text{so} \quad x^2 = 4y^2.$$

Together with $x^2 + 4y^2 = 1$, this gives

$$y = \pm \frac{1}{2\sqrt{2}}, \quad x = \pm \frac{1}{\sqrt{2}}.$$

When $x = 2y$, the points and multiplier are

$$\left(\frac{1}{\sqrt{2}}, \frac{1}{2\sqrt{2}} \right), \left(-\frac{1}{\sqrt{2}}, -\frac{1}{2\sqrt{2}} \right), \quad \lambda = -\frac{e^{-1/4}}{4}.$$

When $x = -2y$, they are

$$\left(\frac{1}{\sqrt{2}}, -\frac{1}{2\sqrt{2}} \right), \left(-\frac{1}{\sqrt{2}}, \frac{1}{2\sqrt{2}} \right), \quad \lambda = \frac{e^{1/4}}{4}.$$

Hence the boundary candidates are

$$\boxed{\left(\pm \frac{1}{\sqrt{2}}, \pm \frac{1}{2\sqrt{2}} \right) \text{ with matched signs, and } \left(\pm \frac{1}{\sqrt{2}}, \mp \frac{1}{2\sqrt{2}} \right)}.$$

(d) The function $f(x, y) = e^{-xy}$ is continuous on $\mathbb{R}^2$. The set C , defined by $x^2 + 4y^2 \leq 1$, is closed and bounded, hence compact. By the Extreme Value Theorem,

$$\boxed{f|_C \text{ attains both an absolute minimum and an absolute maximum.}}$$

(e) At the interior critical point, $f(0, 0) = 1$. At the boundary candidates,

$$xy = \frac{1}{4} \implies f = e^{-1/4}, \quad xy = -\frac{1}{4} \implies f = e^{1/4}.$$

Comparing these values gives

$$\boxed{\min_C f = e^{-1/4} \text{ at } \left(\frac{1}{\sqrt{2}}, \frac{1}{2\sqrt{2}} \right), \left(-\frac{1}{\sqrt{2}}, -\frac{1}{2\sqrt{2}} \right),}$$

$$\boxed{\max_C f = e^{1/4} \text{ at } \left(\frac{1}{\sqrt{2}}, -\frac{1}{2\sqrt{2}} \right), \left(-\frac{1}{\sqrt{2}}, \frac{1}{2\sqrt{2}} \right).}$$

Marking scheme (25 marks)**(a) (5 marks)**

- 3 marks: differentiates and solves $\text{grad } f = 0$.
- 2 marks: identifies $(0, 0)$ as the sole interior critical point.

(b) (5 marks)

- 3 marks: computes the required second derivatives at $(0, 0)$.
- 1 mark: obtains discriminant -1 .
- 1 mark: concludes that the point is a saddle point.

(c) (10 marks)

- 3 marks: states the Lagrange system and boundary constraint.
- 3 marks: derives $x^2 = 4y^2$.
- 3 marks: finds all four boundary points.
- 1 mark: gives the associated function values or multipliers correctly.

(d) (3 marks)

- 1 mark: states continuity of f .
- 1 mark: states compactness of C .
- 1 mark: invokes the Extreme Value Theorem.

(e) (2 marks)

- 1 mark: identifies the absolute minimum and its points.
- 1 mark: identifies the absolute maximum and its points.

Question 4**(25 Marks)****PROBLEM 4.**

(a) Evaluate

$$\int_0^1 \left[\int_0^{\sqrt{(1-x)/3}} e^{y-y^3} dy \right] dx. \quad (10 \text{ points})$$

Hint: Perform a change of order of integration.

(b) Evaluate

$$\int_0^2 \left[\int_0^{\sqrt{2y-y^2}} \sqrt{x^2+y^2} dx \right] dy. \quad (15 \text{ points})$$

Hint: Perform a transformation from Cartesian to polar coordinates.**Solution****Method 1: Change of order and polar coordinates**

(a) The original region is

$$R = \left\{ (x, y) : 0 \leq x \leq 1, 0 \leq y \leq \sqrt{\frac{1-x}{3}} \right\}.$$

Equivalently, it is the first-quadrant region

$$0 \leq y \leq \frac{1}{\sqrt{3}}, \quad 0 \leq x \leq 1 - 3y^2.$$

Changing the order of integration therefore gives

$$\int_0^1 \int_0^{\sqrt{(1-x)/3}} e^{y-y^3} dy dx = \int_0^{1/\sqrt{3}} \int_0^{1-3y^2} e^{y-y^3} dx dy.$$

Integrating first with respect to x ,

$$= \int_0^{1/\sqrt{3}} (1 - 3y^2) e^{y-y^3} dy = \left[e^{y-y^3} \right]_0^{1/\sqrt{3}}.$$

Since $(1/\sqrt{3})^3 = 1/(3\sqrt{3})$,

$$\boxed{e^{2/(3\sqrt{3})} - 1.}$$

(b) The Cartesian bounds describe the right half of the disk

$$x^2 + (y-1)^2 \leq 1, \quad x \geq 0.$$

With $x = r \cos \theta$ and $y = r \sin \theta$, its circular boundary becomes

$$r^2 - 2r \sin \theta = 0,$$

so the polar bounds are

$$0 \leq \theta \leq \frac{\pi}{2}, \quad 0 \leq r \leq 2 \sin \theta.$$

Because $\sqrt{x^2 + y^2} = r$ and $dA = r dr d\theta$,

$$\int_0^2 \int_0^{\sqrt{2y-y^2}} \sqrt{x^2 + y^2} dx dy = \int_0^{\pi/2} \int_0^{2\sin\theta} r^2 dr d\theta.$$

Thus

$$\int_0^{\pi/2} \frac{(2\sin\theta)^3}{3} d\theta = \frac{8}{3} \int_0^{\pi/2} \sin^3\theta d\theta = \frac{8}{3} \cdot \frac{2}{3}.$$

Hence

$$\boxed{\frac{16}{9}}.$$

Marking scheme (25 marks)

(a) (10 marks)

- 3 marks: describes the region correctly after reversing the order.
- 2 marks: supplies the correct reversed bounds.
- 3 marks: integrates with respect to x and recognizes the substitution $u = y - y^3$.
- 2 marks: evaluates the endpoints correctly.

(b) (15 marks)

- 3 marks: recognizes and states the Cartesian disk region.
- 4 marks: obtains the correct polar bounds $0 \leq \theta \leq \pi/2$, $0 \leq r \leq 2\sin\theta$.
- 3 marks: transforms both the integrand and area element correctly.
- 3 marks: evaluates the r -integral and $\int_0^{\pi/2} \sin^3\theta d\theta$.
- 2 marks: obtains $16/9$.